



1/5" 5Mega CMOS Image Sensor

BF2557CS

Datasheet



Contents

1. General Description	3
2. Features	3
3. Applications	3
4. Technical Specifications	4
5. Functional Overview	4
5.1. Pixel Array	5
5.2. Lens Chief Ray Angle (CRA)	6
5.3. Column CDS	6
5.4. Sensor Control	7
5.5. A/D Converter	7
5.6. Automatic Black Control	7
5.7. Analog Gain Control	7
5.8. Exposure Time Control	7
5.9. Line Length	7
5.10. Frame Length	7
5.11. Windowing Control	7
5.12. Mirror/Flip Control	8
6. Specifications	8
6.1. Electrical Characteristics	8
6.1.1. Absolute Maximum Ratings	8
6.1.2. DC Parameters	8
6.2. Electro-Optical Characteristics	9
6.3. MIPI Output Timing	9
6.4. Color Filter Spectral Characteristics	10
7. Two-wire Serial Interface& Register	11
7.1. Theory of Operation	11
7.2. Two-wire Serial Interface Functional Description	12
7.3. The Two-wire Serial Interface Timing	13
7.4. Register Summary	13
7.4.1. Sensor Control Registers	13
7.4.2. Black Control Registers	14
7.4.3. Gain/Exposure Control Registers	16
7.4.4. Binning/BP Registers	16
7.4.5. Test Pattern Registers	17
7.4.6. MIPI Control Registers	17
7.4.7. Windowing Control Registers	19
7.4.8. 7.4.8 OTP Registers	19
7.4.9. System Control Registers	20
8. Package Specification	22
8.1. CSP Package	22
8.2. Pin Descriptions	23
9. Application Timing	24



1. General Description

The BF2557CS is a 1/5-inch 5Mega CMOS Image Sensor which includes CMOS image sensor (CIS) and Sensor control. It is fabricated with the world's most advanced CMOS image sensor process to realize ultra-low dark noise, high sensitivity, high dynamic range and very low power imaging system. The sensor consists of a 2592×1944 pixel array which has an optical format of 1/5 inch. It has integrated noise canceling CDS (Correlated Double Sampling) circuits, analog global gain control, auto black level compensation, on-chip 10-bit ADC. It provides Bayer RGB data with serial MIPI interface. It has a commonly used two-wire serial interface for host to control the operation of the whole sensor.

The product is capable of operating at up to 30 frames per second in 5Mega mode, with complete user control over image. All required sensor control functions, including Exposure control, Gain control, image size and so on, could be programmable through the two-wire serial bus.

2. Features

- 1/5 inch optical format
- 1.12um x1.12um BSI Pixel
- 30 frames/sec @5Mega
- 30 frames/sec @1080P
- 60 frames/sec @720P
- Ultra-low dark noise at high temperature
- Ultra-Low power consumption
- Output formats: Raw Bayer (10bit)
- Output interface: MIPI 2-lane interface
- Horizontal /Vertical mirror
- Auto black level control
- Black sun cancellation, Window control, Exposure/Gain Control
- OTP support (8K bytes)
- 2 x 2 Binning

3. Applications

- Cellular Phone Cameras
- Notebook PC cameras and IP cameras
- PDAs
- MP4
- Digital still cameras and camcorders
- Video telephony and conferencing equipment
- Industrial and environmental systems

4. Technical Specifications

- Optical Format 1/5 inch
- Active pixel array 2592 x 1944
- Pixel size 1.12um x 1.12um
- Sensitivity 0.43 V/Lux-sec
- Dark current 0.2 mV/s at 25°C
- Power supply VDDIO:1.7~3.1V
VDD3A/VDD3C:2.6~3.1V
VDDD:1.15~1.35V
- Power consumption TBD
- Standby current TBD
- SNR(Max) 37.5 dB
- Dynamic range 64 dB
- Operating temperature -20~70°C
- Stable Image temperature 0~60°C
- Lens chief ray angle 31.9° (Non-linear)
- Input clock frequency 6~27MH
- Package type CSP

5. Functional Overview

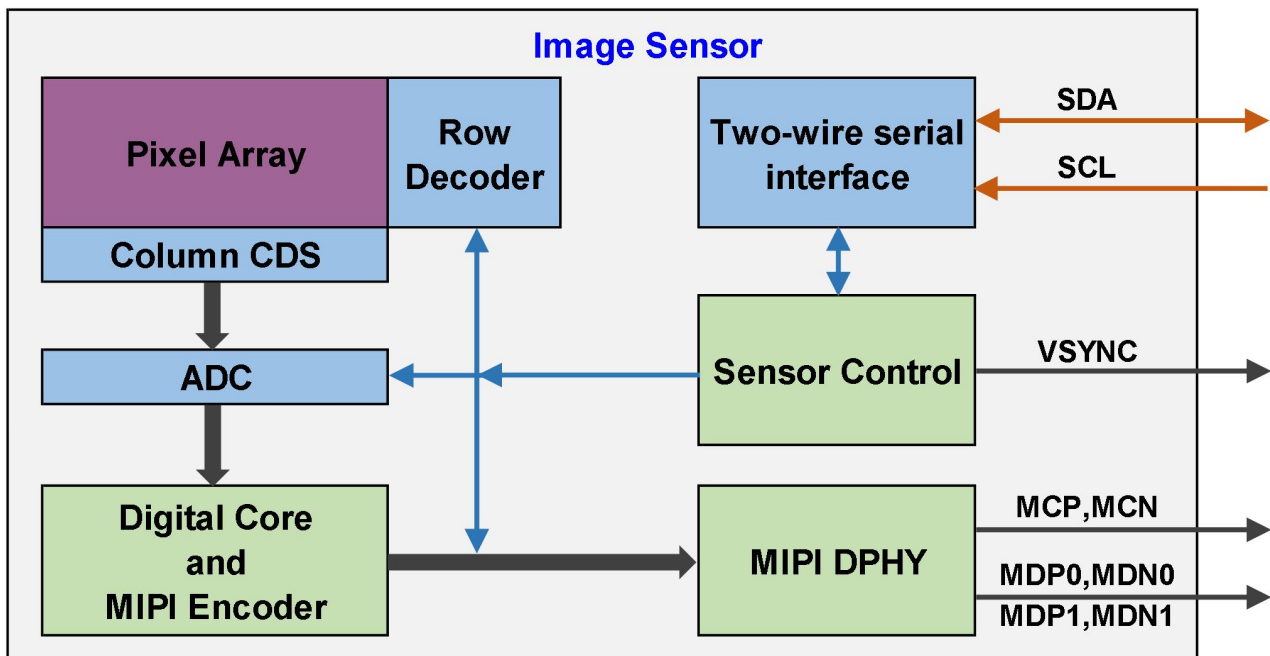


Figure 1 Block Diagram

BF2557CS has an active image array of 2592x1944 pixels. The active pixels are read out progressively through column/row driver circuits. In order to reduce fixed pattern noise, CDS circuits are adopted. The analog signal is transferred to digital signal by A/D converter. It provides global gain to get accurate exposure under different light condition. The digital signals are controlled by Sensor Control block, including Exposure

control, Gain control, Test pattern, Image size control and so on.

BF2557CS has on-chip PLL, it can be used by via two-wire serial interface bus setting.

5.1. Pixel Array

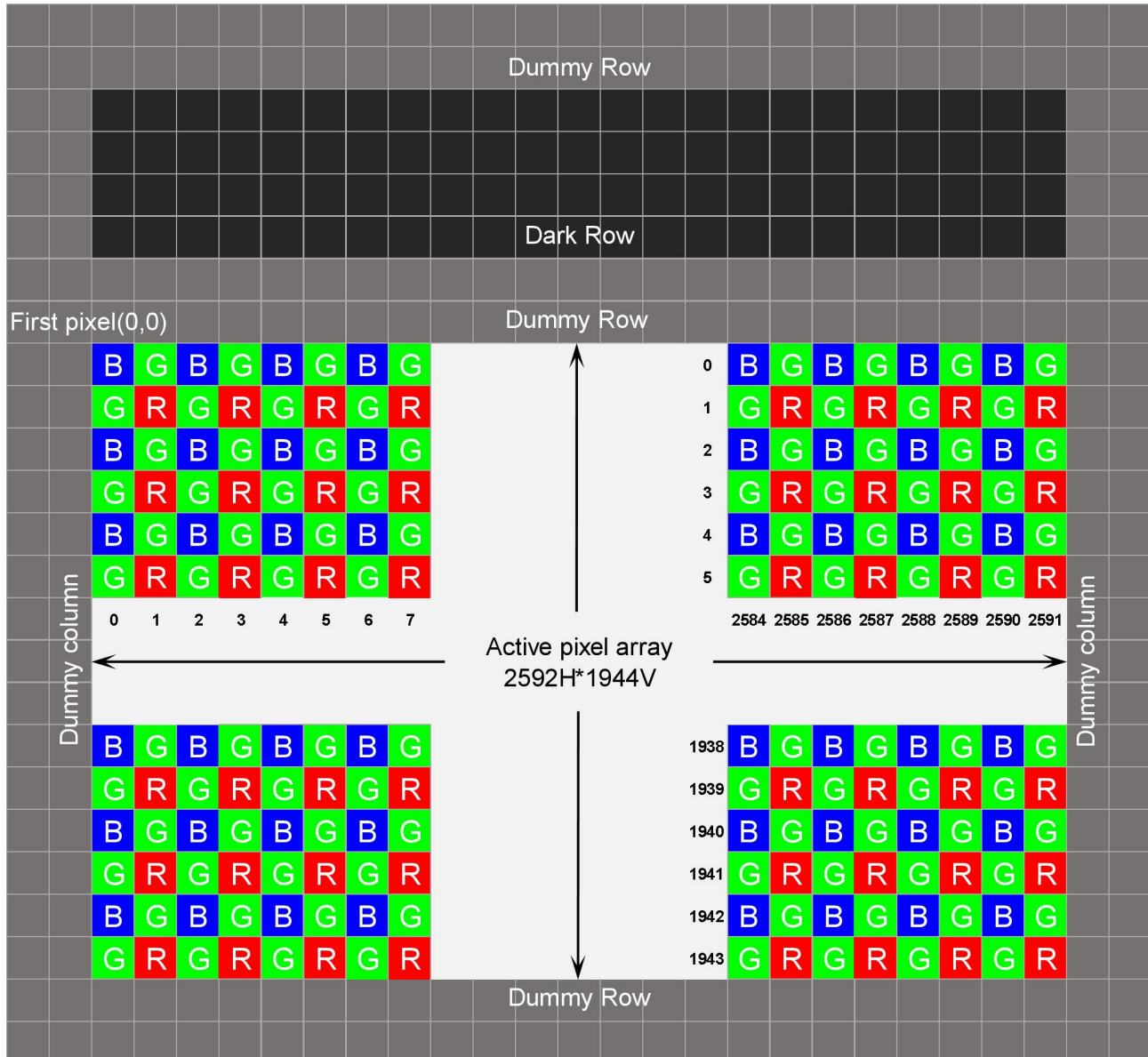


Figure 2 Sensor Array Region

The pixel array includes 2592×1944 effective pixels for imaging, whose address range is from (0, 0) to (2591, 1943). In order to improve the image uniformity, there are extra dummy rows and dummy columns around active array. And dark rows at the top side are for black level control.

Pixel array is covered by Bayer color filters as can be seen in the figure2. The primary color BG/GR array is arranged in line-alternating fashion. Since each pixel can have only one type of color filter on it, only one color component can be obtained by a pixel. BF2557CS can provide the Raw Bayer data through an 10-bit output data bus.

Pixel array output signal order is:

BGBGBG.....

GRGRGR.....

5.2. Lens Chief Ray Angle (CRA)

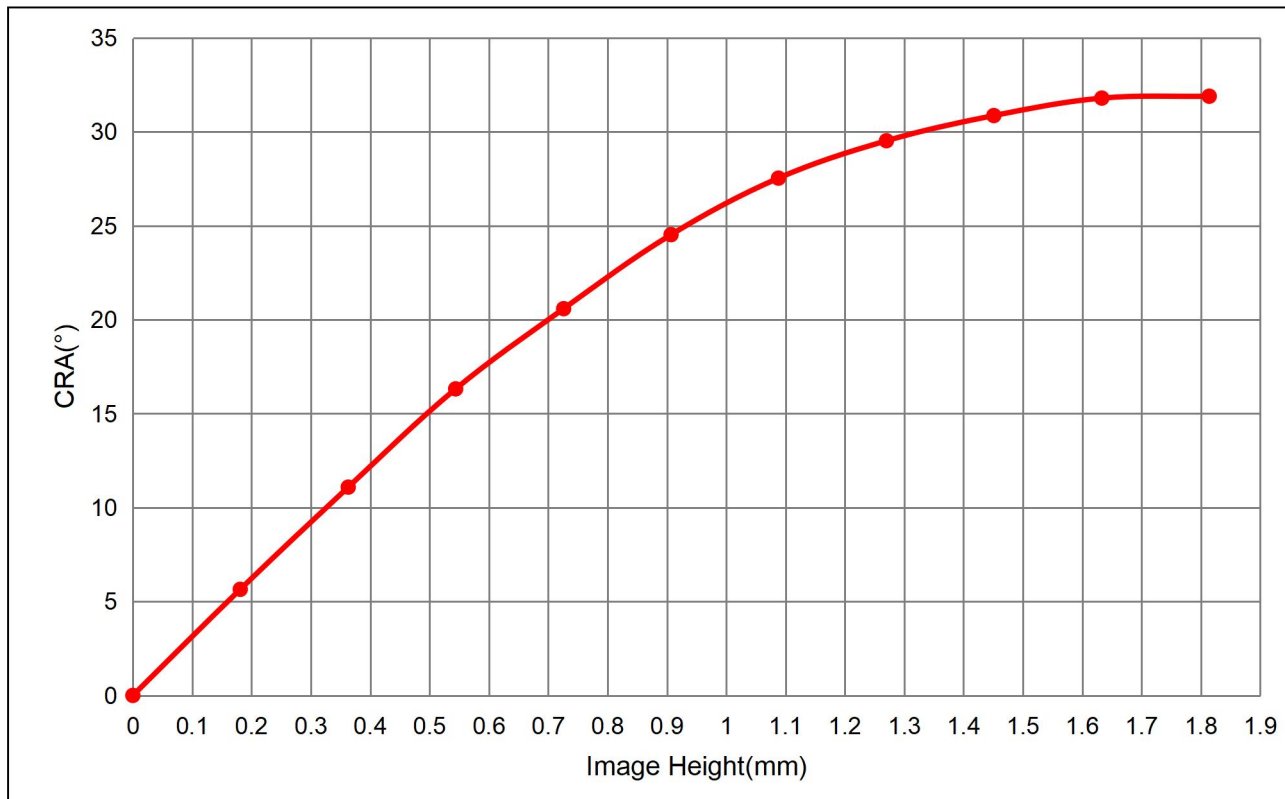


Figure 3 CRA information

Table 1 CRA versus image height plot

Field(%)	Image Height(mm)	CRA(degree)
0	0	0
0.1	0.181	5.65
0.2	0.363	11.09
0.3	0.544	16.32
0.4	0.726	20.59
0.5	0.907	24.53
0.6	1.088	27.54
0.7	1.27	29.53
0.8	1.451	30.87
0.9	1.633	31.8
1	1.814	31.9

5.3. Column CDS

BF2557CS has column/row driver circuits to read out the pixel data progressively. The CDS (Correlated Double Sampling) circuit reduces temporal noise and pixel level FPN (Fixed Pattern Noise). The unique



patented column buffer and ADC circuit remove column level FPN caused by various sources of manufacturing process variations.

5.4. Sensor Control

- Array control and frame generation
- Internal timing signal generation and distribution
- Exposure control, Gain control, Test pattern,
- Image size control
- Frame rate timing
- External timing outputs

5.5.A/D Converter

The analog signals are converted to digital forms and data are streamed out column by column. BF2557CS provides the 10-bit Raw Bayer data through an internal 10-bit data bus.

5.6. Automatic Black Control

The automatic black level control calculates the data of the dark row and controls the lowest black level for output image data.

5.7. Analog Gain Control

The analog gain is linear. The register (0x6a) sets the analog gain. The range of analog gain register is 0x0f-0xff. The maximum analog gain is 16X. The analog gain can be calculated by following formula:

Analog gain=(register value+1)/16.

5.8. Exposure Time Control

BF2557CS supports manual exposure mode. The exposure time can be set by register {0x6b, 0x6c}. The exposure control of BF2557CS is “rolling shutter”.

$$T_{\text{EXPOSURE}} = \text{Exp_TIM}[15:0] \times (\text{Line Length}) \times T_{\text{PCLK}}$$

5.9. Line Length

The line length of BF2557CS is equal to {0x2c[3:0], 0x0b[7:0]}*2.

5.10. Frame Length

The frame length can be read out from register {0x29,0x28}.

The minimum frame length=1970+ register {0x07,0x06}.

If the exposure time is $\leq$ (the minimum frame length-2), the actual frame length is the minimum frame length.

If the exposure time is $>$ (the minimum frame length-2), the actual frame length is equal to the exposure time.

5.11. Windowing Control

User can get image of any size from any location by windowing, which is controlled by registers: 0xca, 0xcb, 0xcc, 0xcd, 0xce, 0xcf.

Horizontal starting address is set by register {0xca[3:0],0xcc};

Horizontal end address is set by register {0xca[7:4],0xcd};

Vertical starting address is set by register {0xcb[2:0],0xce};

Vertical end address is set by register {0xcb[6:4],0xcf};

5.12. Mirror/Flip Control

The image direction can be set by register 0xf2.

Function	Register Address	Register Value	First pixel
Normal	0xf2[1:0]	2'b00	B
Horizontal Mirror	0xf2[1:0]	2'b10	GB
Vertical Flip	0xf2[1:0]	2'b01	GR
Horizontal Mirror&Vertical Flip	0xf2[1:0]	2'b11	R

6. Specifications

6.1. Electrical Characteristics

6.1.1. Absolute Maximum Ratings

- Supply voltage (VDDIO): 1.7 V~3.1 V
- Supply voltage (VDD3A/VDD3C): 2.6V~3.1 V
- Supply voltage (VDDD): 1.15V~1.35 V
- Operating temperature: -20~70°C
- Stable Image temperature: 0~60°C
- ESD Rating, Human Body mode: 2000 V

Caution: Stresses exceeding the absolute maximum ratings may induce failure.

6.1.2. DC Parameters

Table 2 DC Operation Conditions

Symbol	Parameter	Unit	Min.	Typ.	Max.	Notes
VDDIO	I/O power supply	V	1.7	1.8	3.1	1
VDD3A/VDD3C	Analog power supply	V	2.6	2.8	3.1	--
VDDD	Core power supply	V	1.15	1.2	1.35	2
Vih	Input voltage logic "1"	V	0.7*VDDIO	--	--	--
Vil	Input voltage logic "0"	V	--	--	0.2*VDDIO	--
Voh	Output voltage logic "1"	V	0.9*VDDIO	--	--	--

Table 3 AC Operation Conditions

Symbol	Parameter	Unit	Min.	Typ.	Max.	Notes
XCLK	External clock frequency	MHz	--	6~27	--	1
PCLK	Pixel clock of the system	MHz	--	85.2	--	2
SCLK	two-wire serial interface clock frequency	KHz	--	400	--	3

Notes:

- XCLK is the input clock and it is the input of PLL.
- PCLK is the master clock of the system, and it can be generated by PLL.
- SCLK is driven by host processor. For the detail serial bus timing, refer to two-wire serial Interface section.

6.2. Electro-Optical Characteristics

Clock frequency: 85.2MHz.

Operating voltage: VDDIO=1.8V, VDDD=1.2V, VDD3A/VDD3C=2.8V.

Operating temperature: 25°C

Table 4 Electro-Optical Characteristics

Parameter	Unit	Min.	Typ.	Max.	Notes
Sensitivity	V/Lux·sec	--	0.43		1
Dark current	mV/sec	--	0.2	--	2
S/N ratio	dB	--	37.5	--	--
Dynamic Range	dB	--	64	--	--
Frame Rate	fps	--	30	30	3

Notes:

1. With color filter, measured at 50 lux green light condition at room temperature.
2. Measured at dark condition for exposure time of 1s (25 Celsius).
3. 2592×1944 size at PCLK 85.2MHz.

6.3. MIPI Output Timing

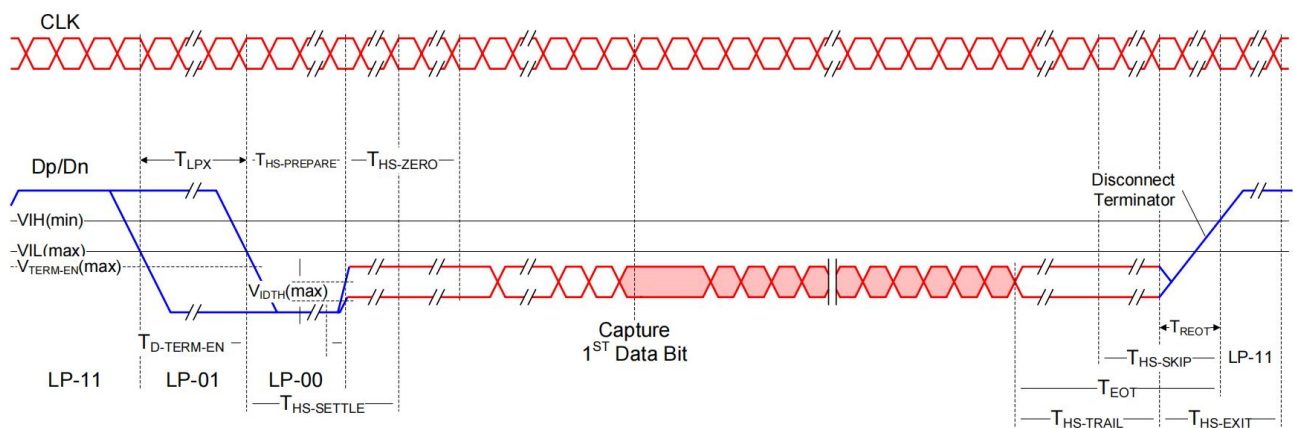


Figure 4 High-Speed Data Transmission in Bursts

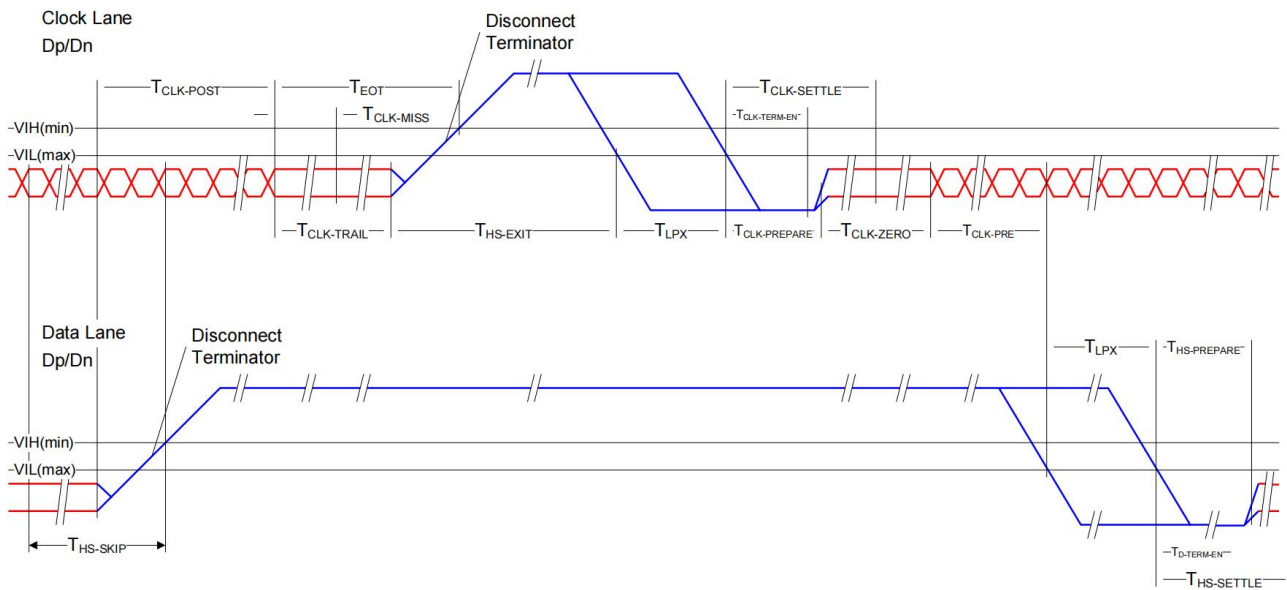


Figure 5 Switching the Clock Lane between Clock Transmission and Low-Power Mode

6.4. Color Filter Spectral Characteristics

The optical spectrum of color filters is shown below.

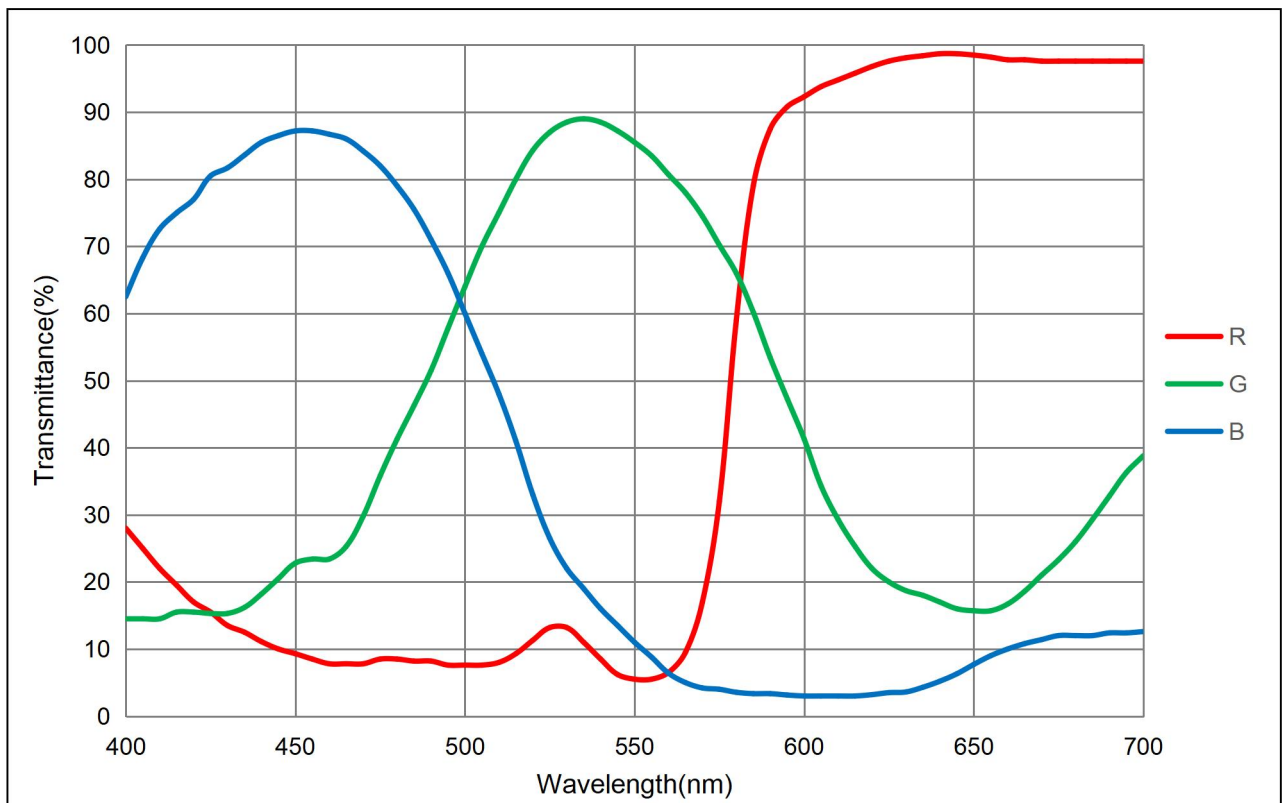


Figure 6 Spectral Characteristics

7. Two-wire Serial Interface& Register

7.1. Theory of Operation

The registers of BF2557CS are written and read through the two-wire serial interface. BF2557CS has two-wire serial interface slave. BF2557CS is controlled by the two-wire serial interface clock (SCLK), which is driven by the two-wire serial interface master. Data is transferred into and out of BF2557CS through the two-wire serial interface data (SDA) line. The SCL and SDA lines are pulled up to VDDIO by a 2kΩ off-chip resistor. Either the slave or the master device can pull the lines down. The two-wire serial interface protocol determines which device is allowed to pull the two lines down at any given time.

Start bit

The start bit is defined as a HIGH to LOW transition of the data line while the clock line is HIGH.

Stop bit

The stop bit is defined as a LOW to HIGH transition of the data line while the clock line is HIGH.

Slave Address

The 8-bit address of a two-wire serial interface device consists of 7 bits of address and 1 bit of direction. A0 in the LSB of the address indicates write mode, and A1 indicates read mode.

Data bit transfer

One data bit is transferred during each clock pulse. The two-wire serial interface clock pulse is provided by the master. The data must be stable during the HIGH period of the two-wire serial interface clock: it can only change when the two-wire serial interface clock is LOW. Data is transferred 8 bits at a time, followed by an acknowledge bit.

Acknowledge bit

The receiver generates the acknowledge clock pulse. The transmitter (which is the master when writing, or the slave when reading) releases the data line, and receiver indicates an acknowledge bit by pulling the data line low during the acknowledge clock pulse.

No-acknowledge bit

The no-acknowledge bit is generated when the data line is not pulled down by the receiver during the acknowledge clock pulse. A no-acknowledge bit is used to terminate a read sequence.

Sequence

A typical read or write sequence begins by the master sending a start bit. After start bit, the master sends the slave device's 8-bit address. The last bit of the address determines if the request will be a read or a write, where a 0 indicates a write and a 1 indicates a read. The slave device acknowledges its address by sending an acknowledge bit back to the master.

If the request was a write, the master then transfers the 8-bit register address to which a write should take place. The slave sends an acknowledge bit to indicate that the register address has been received. The master then transfers the data 8 bits at a time, with the slave sending an acknowledge bit after each 8 bits. The BF2557CS uses 8 bit data for its internal registers, thus requiring one 8-bit transfer to write to one register. After 8 bits are transferred, the register address is automatically incremented, so that the next 8 bits are written to the next register address. The master stops writing by sending a start or stop bit.

A typical read sequence is executed as follows. First the master sends the write-mode slave address and 8-bit register address just as in the write request. The master then sends a start bit and the read-mode slave

address. The master then clocks out the register data 8 bits at a time. The master sends an acknowledge bit after each 8-bit transfer. The register address is auto-incremented after each 8 bit is transferred. The data transfer is stopped when the master sends a no-acknowledge bit.

7.2. Two-wire Serial Interface Functional Description

BF2557CS Device Address:

Table 5 Device Address

ID_SEL	Slave address write mode	Slave address read mode
Low(default)	0x7c	0x7d
High	0xdc	0xdd

Single Write Mode Operation

S	Slave address	W	A	Register address	A	Data	A	P
---	---------------	---	---	------------------	---	------	---	---

Multiple Write Mode (Register address is increased automatically)¹ Operation

Example Write mode (Register address is increased automatically) Operation										
S	Slave address		W	A	Register address		A	Data	A	...
...			A	Data	A	Data	A	...		
							A	Data	A	P

Single Read Mode Operation

S	Slave address	W	A	Register address	A	...
Sr	Slave address	R	A	Data	NA	P

Multiple Read Mode (Register address is increased automatically)¹ Operation

S	Slave address	W	A	Register address	A	...		
Sr	Slave address	R	A	Data	A	Data	A	...
...			A	Data	A	Data	NA	P

 From master to slave

 From slave to master

S: Start condition.

Sr: Repeated Start (Start without preceding stop.)

A: Acknowledge bit.

NA: No Acknowledge.

Data: 8-bit data

P: Stop condition

R/W: Read/Write selection. High = read, LOW = write.

Note1: Continuous writing or reading without any interrupt increases the register address automatically. If the address is increased above valid register address range, further writing does not affect the chip operation in write mode. Data from invalid registers are undefined in read mode.

7.3. The Two-wire Serial Interface Timing

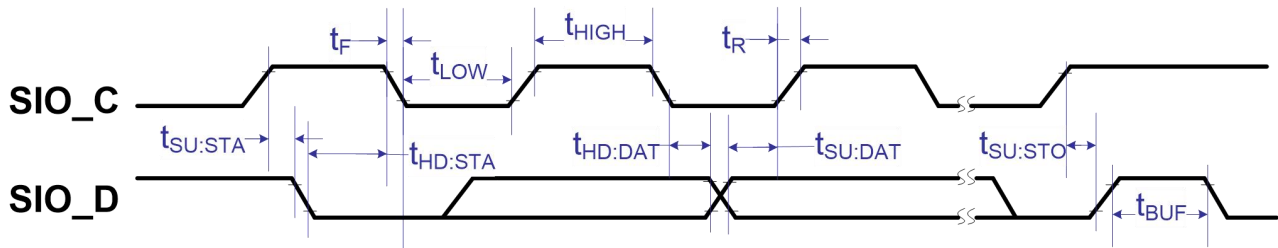


Figure 7 Two-Wire Serial Interface Timing

Table 6 AC Characteristics

Symbol	Descriptions	Min.	Typ.	Max.	Unit
t_R, t_F	two-wire serial interface rise/fall times	--	--	300	ns
t_{LOW}	Clock Low Period	1.3	--	--	us
t_{HIGH}	Clock High Period	600	--	--	ns
$t_{HD:STA}$	Start condition Hold Time	600	--	--	ns
$t_{SU:STA}$	Start condition Setup Time	600	--	--	ns
$t_{HD:DAT}$	Data-in Hold Time	0	--	--	ns
$t_{SU:DAT}$	Data-in Setup Time	100	--	--	ns
$t_{SU:STO}$	Stop condition Setup Time	600	--	--	ns

7.4. Register Summary

7.4.1. Sensor Control Registers

Table 7.4.1 Sensor Control Registers

Address	Name	Width	Default value	Description
0x01	SC_CNTL1	8	0x42	Bit[7]: PD_OUT SEL. Bit[6]: VBLANK SEL. Bit[5]: GLB_GAIN_IN SEL. Bit[4]: VSYNC_MIPI SEL. Bit[3:0]: Reserved.
0x02	SC_CNTL2	8	0x8e	Reserved.
0x03	SC_CNTL3	8	0xc0	Bit[7:6]: Reserved. Bit[5]: 1'B0,FSYNC IO enable. 1'B1,FSYNC IO disable. Bit[4]: 1'B0,Master sync mode. 1'B1,Slave sync mode. Bit[3]: FSIN Sel. Bit[2]: Reserved. Bit[1]: 1'B1,Dynamic mismatch sync mode on; 1'B0,Dynamic mismatch sync mode off; Bit[0]: Dual sync delay Sel.

Address	Name	Width	Default value	Description
0x04	SC_CNTL4	8	0x00	Dual sync offset
0x05	DSYNC_OFFSET[7:0]	8	0x00	The offset of dynamic mismatch
0x06	VBLANK_MIN[7:0]	8	0x00	Dummy line low 8 Bits
0x07	VBLANK_MIN[15:8]	8	0x00	Dummy line high 8 Bits
0x08	HSYST	8	0x70	Control the rising edged of HSYNC, HSYNC rising edge low 8 Bits
0x09	HSYEN	8	0x2e	Control the falling edged of HSYNC, HSYNC falling edge low 8 Bits
0x0a	H_HSYNC_EDGE	8	0x55	Bit[7:4]: HSYNC rising edge[11'b1 8]; Bit[3:0]: HSYNC falling edge[11'b1 8];
0x0b	LINE_LREG[7:0]	8	0x82	Bit[7:0]: LINE_LENGTH[7:0].
0x27	SCAN_IDLE_L INTVL	8	0xf1	Bit[7:4]: SCAN_IDLE_L[3:0] Bit[3]: Reserved Bit[2:0]: INTVL
0x28	FRAME_LENGTH_O UT[7:0]	8	0xb2	Bit[7:0]: FRAME_LENGTH_DEFAULT[7:0].
0x29	FRAME_LENGTH_O UT[15:8]	8	0x07	Bit[7:0]: FRAME_LENGTH_DEFAULT[15:8].
0x2a	RST_COUNTER_F	8	0x00	Bit[7:5]: Reserved Bit[4:0]: Control the falling edge of RST_COUNTER
0x2b	MIPI_STEN	8	0x08	Control VSYNC_MIPI falling edge.
0x2c	LINE_HREG	8	0x05	Bit[3:0] :LINE_LENGTH[11:8].
0x40	DELAY_CNTL	8	0x02	Control the delay of OTP_READY

7.4.2. Black Control Registers

Table 7.4.2 Black Control Registers

Address	Name	Width	Default value	Description
0x50	DARKE0_AVER[10] DARKO0_AVER[10] DARKE1_AVER[10] DARKO1_AVER[10]	4	RO	Bit[7]: Current frame BLC value for E row E col; Bit[6]: Current frame BLC value for E row O col; Bit[5]: Current frame BLC value for O row E col; Bit[4]: Current frame BLC value for O row O col;
0x51	DARKE0_AVER[9:8] DARKO0_AVER[9:8] DARKE1_AVER[9:8] DARKO1_AVER[9:8]	8	RO	Bit[7:6]: Current frame BLC value for E row E col; Bit[5:4]: Current frame BLC value for E row O col; Bit[3:2]: Current frame BLC value for O row E col; Bit[1'b1 0]: Current frame BLC value for O row O col;
0x52	DARKE0_AVER	8	RO	Current frame BLC value for E row E col;
0x53	DARKO0_AVER	8	RO	Current frame BLC value for E row O col;
0x54	DARKE1_AVER	8	RO	Current frame BLC value for O row E col;
0x55	DARKO1_AVER	8	RO	Current frame BLC value for O row O col;

Address	Name	Width	Default value	Description
0x56	AVER_TAR_E0[9:8] AVER_TAR_O0[9:8] AVER_TAR_E1[9:8] AVER_TAR_O1[9:8]	8	0x00	Bit[7:6]: Black level target for even col Bit[5:4]: Black level target for odd col Bit[3:2]: Black level target for even col Bit[1:0]: Black level target for odd col
0x57	AVER_TAR_E0	8	0x00	Black level target for E row E col;
0x58	AVER_TAR_O0	8	0x00	Black level target for E row O col;
0x59	AVER_TAR_E1	8	0x00	Black level target for O row E col;
0x5a	AVER_TAR_O1	8	0x00	Black level target for O row O col;
0x5b	AVER_E0[10] AVER_O0[10] AVER_E1[10] AVER_O1[10]	4	RO	Bit[7]: Read out black level for even col and even row. Bit[6]: Read out black level for odd col and even row. Bit[5]: Read out black level for even col and odd row. Bit[4]: Read out black level for odd col and odd row.
0x5c	AVER_E0[9:8] AVER_O0[9:8] AVER_E1[9:8] AVER_O1[9:8]	8	RO	Bit[7:6]: Read out black level for even col and even row. Bit[5:4]: Read out black level for odd col and even row. Bit[3:2]: Read out black level for even col and odd row. Bit[1'b1 0]: Read out black level for odd col and odd row.
0x5d	AVER_E0	8	RO	Read out black aver for E row E col,this value has been adjusted
0x5e	AVER_O0	8	RO	Read out black aver for E row O col,this value has been adjusted
0x5f	AVER_E1	8	RO	Read out black aver for O row E col,this value has been adjusted
0x60	AVER_O1	8	RO	Read out black aver for O row O col,this value has been adjusted
0x61	COEF	8	0x80	Black aver adjust coef.
0x62	MODE_CNTL1	8	0x5f	Bit[7] : TEST_PATTERN 1'B1,Use fixed value for test 1'B0,Close test_pattern Bit[6]: SEL_LIMIT_WIDE Bit[4]: LAST_EN Bit[3:2]: DARKROW_BIN_SEL: binning mode dark Bit[1]: DARKROW SEL.
0x63	MODE_CNTL	8	0x23	Bit[7]: BYPASS_BC. Bit[6:4]: DARKROW_SEL. Bit[3]: DGAD_MODE. Bit[1]: MODE_CTL. Bit[0]: DIG_MODE_CTL.
0x91	AVER_LOCK	8	0x21	Lock value to update with current frame black level.
0x92	AVER_LOCK_SUB	2	0x00	AVER_LOCK_SUB
0x93	FRAME_AVER_TH	8	0x10	Used as black level update threshold

7.4.3. Gain/Exposure Control Registers

Table 7.4.3 Gain/Exposure Control Registers

Address	Name	Width	Default value	Description
0x8b	DGCTRL	4	0x00	Bit[7:4]: Reserved; Bit[3]: Color Gain control: 1'B1,Bypass 1'B0,Normal Bit[2]: Dig Gain control: 1'B1,Bypass 1'B0,Normal Bit[1]: OFFSET_COLOR_M Bit[0]: OFFSET_DIG_M
0x8c	OFFSET_COLOR_ME	8	0x00	bit[7:0]: The manual written color gain offset.
0x8d	OFFSET_COLOR_MO	8	0x00	bit[7:0]: The manual written color lgain offset.
0x8e	OFFSET_DIG_ME	8	0x00	bit[7:0]: The manual written dig gain offset.
0x8f	OFFSET_DIG_MO	8	0x00	bit[7:0]: The manual written dig gain offset.
0x90	COMP_GAIN	6	0x00	bit[7:6]: Reserved. bit[5:0]: COMP_GAIN,LSB control.
0x64	Y_AVER	8	RO	The Y_aver of the current frame
0x65	UPDATE_EN	1	0x1h	Bit[7]: Global switch. 1'B0,On 1'B1,Off.
0x66	R_GAIN	8	0x80	Red gain.
0x67	B_GAIN	8	0x80	Blue gain.
0x68	G0_GAIN	8	0x80	Green0(B) gain.
0x69	G1_GAIN	8	0x80	Green1(R) gain.
0x6a	GLB_GAIN	8	0x20	Global gain.
0x6b	INT_TIME_H	8	0x02	Integration time MSB.
0x6c	INT_TIME_L	8	0xa2	Integration time LSB.
0x6d	GAIN_L	8	0x00	The color gain LSB.
0x6e	DIG_GAIN_H	3	0x00	The value of DIG_GAIN MSB
0x6f	DIG_GAIN_L	8	0x40	The value of DIG_GAIN LSB

7.4.4. Binning/BP Registers

Table 7.4.4 Binning/BP Registers

Address	Name	Width	Default value	Description
0x00	SC_CNTL0	8	0x10	Bit[7:4]: Reserved Bit[3]: Binning enable. Bit[2]: Binning mode. Bit[1: 0]: Reserved.

Address	Name	Width	Default value	Description
0x98	BP_COM	8	0x80	Bit[7]: BP_ENABLE 1'B0,Bp disable 1'B1,Bp enable Bit[6:5]: Reserved; Bit[4]: DAT_EXCH 1'B0,Do not exchange 1'B1,Do data exchange Bit[3:0]: Reserved;
0xa0	BINNING_SEL	8	0x00	Bit[7:3]: Reserved; Bit[1]: 1'B1,Double luminance 1'B0,Normal Bit[1]: 1'B1,Daul mode 1'B0,Tri mode

7.4.5. Test Pattern Registers

Table 7.4.5 Test Pattern Registers

Address	Name	Width	Default value	Description
0x94	TEST_MODE	8	0x00	Bit[7]: TEST_MODE_EN: Bit[6:0]: Test mode
0x95	MAN_R	8	0x80	Define R value.
0x96	MAN_G	8	0x80	Define G value.
0x97	MAN_B	8	0x80	Define B value.

7.4.6. MIPI Control Registers

Table 7.4.6 MIPI Control Registers

Address	Name	Width	Default value	Description
0x70	D_LPX	8	0x08	Bit[7:0]: The length of LPX
0x71	HS_PREPARE	8	0x07	Bit[7:0]: The length of HS_PREPARE
0x72	HS_ZERO	8	0x16	Bit[7:0]: The length of HS_ZERO
0x73	HS_TRAIL	8	0x09	Bit[7:0]: The length of HS_TRAIL
0x74	Clk_LPX	8	0x08	Bit[7:0]: The length of Clk_LPX
0x75	Clk_PREPARE	8	0x06	Bit[7:0]: The length of Clk_PREPARE
0x76	Clk_ZERO	8	0x30	Bit[7:0]: The length of Clk_ZERO
0x77	Clk_PRE	8	0x02	Bit[7:0]: The length of Clk_PRE
0x78	Clk_POST	8	0x10	Bit[7:0]: The length of Clk_POST
0x79	Clk_TRAIL	8	0x0a	Bit[7:0]: The length of Clk_TRAIL
0x7a	MIPI_CNTL0	8	0x2b	Bit[7:6]: VC,Virtual Channel Identifier;default:00b. Bit[5:0]: Image_type
0x7b	frame_num_max_H	8	0x00	High bytes of Max Frame_Counter of Frame Sync Short Packet
0x7c	frame_num_max_L	8	0x00	Low bytes of Max Frame_Counter of Frame Sync Short

Address	Name	Width	Default value	Description
				Packet
0x7d	MIPI_CNTL1	8	0x1f	Bit[7:5]: Control the ECC Generation. Bit[4]: 1'B1,MIPI enable; 1'B0,MIPI disable. Bit[3]: 1'B1,Data lane 1 enable; 1'B0,Data lane 1 disable. Bit[2]: 1'B1,Data lane 0 enable; 1'B0,Data lane 0 disable. Bit[1]: 1'B1,Clk lane enable; 1'B0,Clk lane disable. Bit[0]: 1'B0,Continuous clock mode; 1'B1,Non-continuous clock mode.
0x7e	MIPI_CNTL2	8	0x90	Bit[7]: 1'B0,1-Lane system; 1'B1,2-Lane system. Bit[6]: Data output sel. Bit[5]: 1'B0,Data lane1 counter generate clk lane trailer; 1'B1,Data lane0 counter generate clk lane trailer; Bit[4]: Raw10 enable. 1'B1,The data is raw10; 1'B0,The data is raw8. Bit[3]: 1'B1,LS and LE will be output ; 1'B0,LS and LE didn't be output. Bit[2]: 1'B0,WC in LS and LE is always 0; 1'B1,WC in LS and LE isn't 0; Bit[1]: 1'B0,WC in FS and FE is always 0; 1'B1,WC in FS and FE isn't 0 . Bit[0]: 1'B0,VSYNC_IMAGE is be packed 1'B1,VSYNC_DAT is be packed.
0x7f	MIPI_CNTL3	8	0x08	Bit[7]: 1'B1,Between line and line,frame and frame,the state of the DP and DN is HS. 1'B0,Between line and line,frame and frame,the state of the DP and DN is LP. Bit[6]: 1'B1,Enable clock lane ULPS state; 1'B0,Disable clock lane ULPS state; Bit[5]: 1'B1,Enable data lane 1 ULPS state; 1'B0,Disable data lane 1 ULPS state; Bit[4]: 1'B1,Enable data lane 0 ULPS state; 1'B0,Disable data lane 0 ULPS state; Bit[3:0]: The length of each individual LP state period.
0x80	MIPI_CNTL4	8	0x3f	Bit[7:6]: FIX_DATA[9:8]; Bit[5:4]: When Clk_Lane_en=1, CLK_LP=CLK_Lane_DIS_LP; Bit[3:2]:When Dat_Lane_en1=1, CLK_LP=Dat_Lane1_DIS_LP; Bit[1'b1 0]: When Dat_Lane_en0=1, CLK_LP=Dat_Lane0_DIS_LP;

Address	Name	Width	Default value	Description
0x81	FIX_data_L	8	0x00	Bit[7:0]: FIX_DATA[7:0];
0x82	MIPI_CNTL5	4	0x01	Reserved.

7.4.7. Windowing Control Registers

Table 7.4.7 Windowing Control Registers

Address	Name	Width	Default value	Description
0xc9	COM1	8	0x00	Bit[7]: Bypass output control or not 1'B1,Bypass; 1'B0,Not bypass Bit[6:4]: LSB select for DATA Bit[3:2]: Output select for VSYNC_OUT Bit[1]: CROSS_CTRL, control data during blank 1'B1,Always has data no matter vsync_dat/HREF=0 or not; 1'B0,No data when vsync_dat/HREF=0; Bit[0]: HREF_MODE 1'B1,There's href during Vblank 1'B0,No href during Vblank
0xca	HWIN	8	0xa0	Bit[7:4]: MSB of HREF end high 4 bits(low 8 bits at HSTOP[7:0]) Bit[3:0]: MSB of HREF start high 4 bits(low 8 bits at HSTART[7:0])
0xcb	VWIN	8	0x70	Bit[7]: Reserved Bit[6:4]: MSB of VREF end high 3 bits(low 8 bits at VSTOP[7:0]) Bit[3]: Reserved Bit[2:0]: MSB of VREF start high 3 bits(low 8 bits at VSTART[7:0])
0xcc	HSTART	8	0x04	Bit[7:0]: Output Format-Horizontal Frame(HREF column)start
0xcd	HSTOP	8	0x24	Bit[7:0]: Output Format-Horizontal Frame(HREF column)end
0xce	VSTART	8	0x04	Bit[7:0]: Output Format-Vertical Frame(row)start
0xcf	VSTOP	8	0x9c	Bit[7:0]: Output Format-Vertical Frame(row)end
0xd0	SKIP_CTR	8	0x00	Bit[7]: DARK_SEL Bit[6:5]: Reserved Bit[4:0]: Skip frame counter

7.4.8. 7.4.8 OTP Registers

Table 7.4.8 OTP Registers

Address	Name	Width	Default value	Description
0x99	EFUSE_ADDRH	8	0x00	Bit[7:5]: Reserved Bit[4:0]: MSB of the EFUSE address
0x9a	EFUSE_ADDRL	8	0x00	Bit[7:0]: LSB of the EFUSE address
0x9b	EFUSE_DIN	8	0x00	Bit[7:0]: EFUSE data to write
0x9c	EFUSE_RD_CTR EFUSE_WE_CTR EFUSE_UPDM EFUSE_RA_H	8	0x00	Bit[7]: EFUSE_RD_CTR: EFUSE read control Bit[6]: EFUSE_WE_CTR: EFUSE write control Bit[5]: EFUSE_UPDM Bit[4]: EFUSE_RA_H Bit[3]: SRAM_DELAY_SEL Bit[2]: SRAM LENG_SEL
0x9d	EFUSE_DOUT_S	8	RO	OTP Read Data
0x9e	TSPPGAVDDD	8	0x00	Bit[7:6]: TEST_MODE Bit[5]: INHIBIT Bit[4]: POWER_EN: EFUSE power Bit[3]: FALL_FLAG: write data fall. read only Bit[2]: BCT_FLAG: BCT result. read only Bit[1]: TESTDEC_MFLAG: TESTDEC main array flag. read only Bit[0]: TESTDEC_RFLAG: TESTDEC redundant array flag. read only
0x9f	OTP_AVTIVE	8	0x15	Bit[7]: ADD_MODE Bit[6]: SRAM_TEST_OKD: read only Bit[5]: OTP_AVTIVE: read only Bit[4]: CEB_SRAM_SEL: Bit[3:0]: Reserved.

7.4.9. System Control Registers

Table 7.4.9 System Control Registers

Address	Name	Width	Default value	Description
0xf2	MV_REGD	2	0x00	Bit[1]: Mirror, 1'B0,Normal image 1'B1,Mirror image; Bit[0]: Vertical Flip, 1'B0,Normal image 1'B1,Vertically flip image;
0xe0	REG_ANA0	8	0x84	VDDP SEL
0xe1	REG_ANA1	8	0xb2	ANA SEL.
0xe2	REG_ANA2	8	0xe2	ANA SEL.
0xe3	REG_ANA3	8	0xa0	DRV SEL.
0xe4	REG_COM0	8	0x00	PD SEL.
0xe5	REG_COM1	8	0x08	CLK SEL.
0xe6	REG_COM2	8	0x02	CLK SEL.
0xe7	REG_COM3	8	0x47	Bit[7]: En_Bin.



Address	Name	Width	Default value	Description
				Bit[6:4]: VRSEL. Bit[3]: Sel Rout of LVDS for DAT&CLK. Bit[2]: Sel Rout of LVDS for DAT1. Bit[1]: Sel Rout of LVDS for DAT0. Bit[0]: Sel Rout of LVDS for CLK.
0xe8	REG_COM4	8	0x20	Bit[7]: VRSEL2. Bit[6:5]: Sel VREF1P2 for LP. Bit[4]: OEN_LP. Bit[3:0]: Sel DAT phase for MIPI.
0xea	REG_COM6	8	0x16	Bit[7:6]: Id Sel. Bit[5:4]: LPF Sel. Bit[3:0]: Sel for Pre-DIV[3:0]
0xeb	REG_COM7	8	0xd0	Bit[7:0]: PLL LOOP DIV[7:0]
0xec	REG_COM8	8	0x00	Bit[7:1]:Reserved. Bit[0]: PLL_LOOP DIV[8].
0xed	REG_COM9	8	0x2c	ANA SEL.
0xee	REG_OTP	8	0x3d	Bit[7:5]: CP_TRIM[2:0]. Bit[4]: VRDAPVIFY_TRIM Bit[3:0]: IPGM_TRIM[3:0]
0xf0	REGF0	1	0x0h	Bit[7:1]:Reserved; Bit[0]: SCCB reset,register reset 1'B0,Off 1'B1,on
0xf1	REGF1	1	0x01	SOFT_PD_REG.
0xf3	REGF3	1	0x01	bit[1]: READ_C_EN,read in serial mode.
0xfb	VER_BME	4	RO	Bit[7:4]:Reserved; Bit[3:0]: Version of product.
0xfc	PIDH_BME	8	RO	Bit[7:0]: BME ID HSB
0xfd	PIDL_BME	8	RO	Bit[7:0]: BME ID LSB

8. Package Specification

8.1. CSP Package

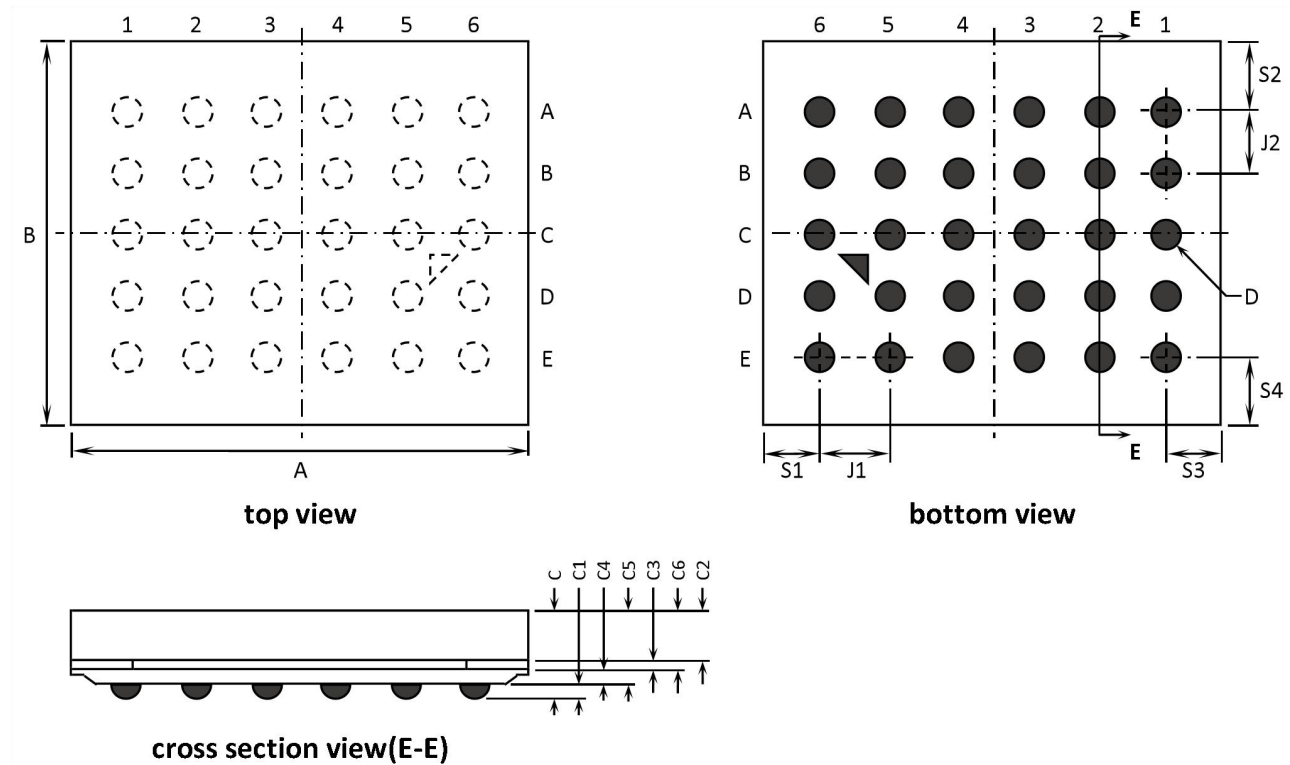


Figure 8 CSP dimension description

Table 8 CSP Dimensions

Description	Symbol	Nominal	Min	Max
		Millimeters		
Package Body Dimension X	A	3.730	3.705	3.755
Package Body Dimension Y	B	3.120	3.095	3.145
Package Height	C	0.620	0.565	0.675
Ball Height	C1	0.120	0.090	0.150
Glass Thickness	C2	0.300	0.290	0.310
Cavity wall height	C3	0.035	0.031	0.039
Si Thickness	C4	0.130	0.120	0.140
Package Body Thickness	C5	0.500	0.465	0.535
Thickness from top glass surface to wafer	C6	0.335	0.330	0.340
Ball Diameter	D	0.230	0.200	0.260
Total Ball Count	N	30		
Pins pitch X axis	J1	0.560	0.550	0.570
Pins pitch Y axis	J2	0.500	0.490	0.510



Edge to Ball Center Distance along X	S1	0.465	0.435	0.495
Edge to Ball Center Distance along Y	S2	0.565	0.535	0.595
Edge to Ball Center Distance along X	S3	0.465	0.435	0.495
Edge to Ball Center Distance along Y	S4	0.555	0.525	0.585

8.2. Pin Descriptions

Table 9 Pin Descriptions

Pin Number	Name	Pin Type	Function/Description
A1	VDD3A	Supply	Analog power supply: 2.6V~3.1V
A2	VDDD	Supply	Digital core power supply: 1.15V~1.35V
A3	ID_SEL	Input	I2C Address Selection. ID_SEL=Low: 0x7c/0x7d; ID_SEL=High: 0xdc/0xdd.
A4	SCL	Input	SCCB serial interface clock input
A5	VDDIO	Supply	I/O power supply: 1.7V~3.1V
A6	VDD3A	Supply	Analog power supply: 2.6V~3.1V
B1	VSSA	Ground	Analog ground
B2	XPD	Input	Power Down Mode Selection. 0: Power down mode; 1: Normal mode.
B3	VSSD	Ground	Digital ground
B4	SDA	I/O	SCCB serial interface data inout
B5	FSYNC	I/O	Frame Sync
B6	VDDD	Supply	Digital core power supply: 1.15V~1.35V
C1	VRP	I/O	Inner analog voltage.
C2	VTN	I/O	Inner analog voltage.
C3	MDN1	I/O	Neg Data lane1 of MIPI
C4	XCLK	Input	System input clock
C5	VSSD	Ground	Digital ground
C6	VSSA	Ground	Analog ground
D1	VDD3C	Supply	Analog power supply: 2.6V~3.1V
D2	VSSD	Ground	Digital ground
D3	MDP1	I/O	Pos Data lane1 of MIPI
D4	MCN	I/O	Neg Clock lane of MIPI
D5	MDN0	I/O	Neg Data lane0 of MIPI
D6	VSSD	Ground	Digital ground
E1	VSSA	Ground	Analog ground
E2	VDD3A	Supply	Analog power supply: 2.6V~3.1V
E3	VDDD	Supply	Digital core power supply: 1.15V~1.35V
E4	MCP	I/O	Pos Clock lane of MIPI
E5	MDP0	I/O	Pos Data lane0 of MIPI

Pin Number	Name	Pin Type	Function/Description
E6	VDDD	Supply	Digital core power supply: 1.15V~1.35V

9. Application Timing

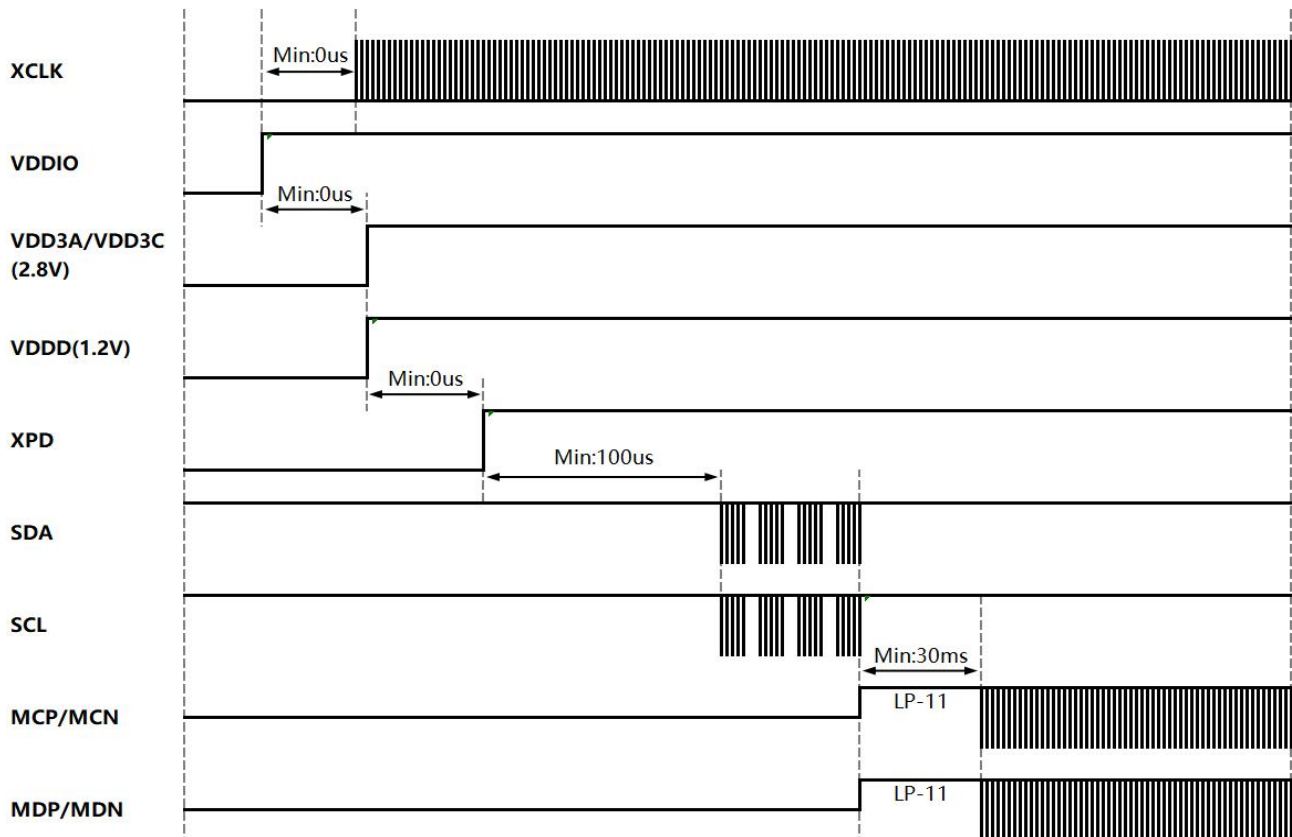


Figure 9 Power on sequence

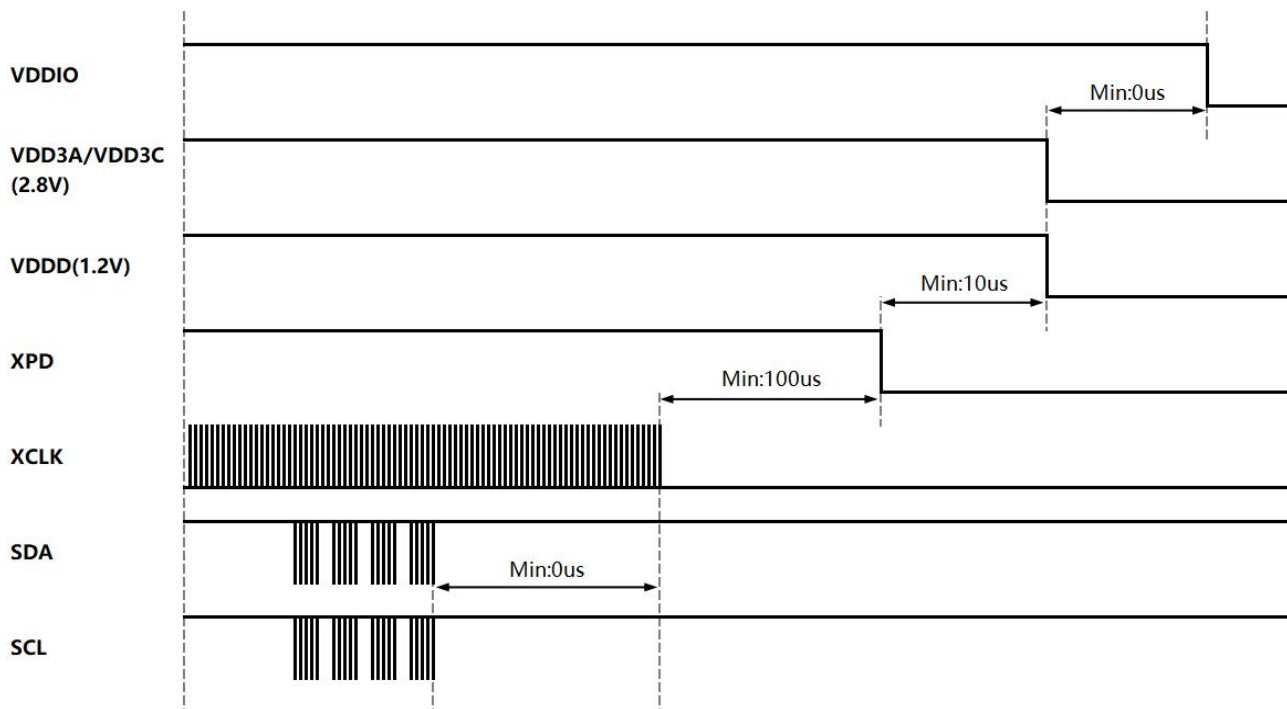


Figure 10 Power off sequence

RESTRICTIONS ON PRODUCT USE

- The information contained herein is subject to change without notice.
- BYD Semiconductor Company Limited exerts the greatest possible effort to ensure high quality and reliability. Nevertheless, semiconductor devices in general can malfunction or fail due to their inherent electrical sensitivity and vulnerability to physical stress. It is the responsibility of the buyer, when utilizing Semiconductor products, to comply with the standards of safety in making a safe design for the entire system, including redundancy, fire-prevention measures, and malfunction prevention, to prevent any accidents, fires, or community damage that may ensue. In developing your designs, please ensure that Semiconductor products are used within specified operating ranges as set forth in the most recent Semiconductor products specifications.
- The Semiconductor products listed in this document are intended for usage in general electronics applications (computer, personal equipment, office equipment, measuring equipment, industrial robotics, domestic appliances, etc.). These Semiconductor products are neither intended nor warranted for usage in equipment that requires extraordinarily high quality and/or reliability or a malfunction or failure of which may cause loss of human life or bodily injury ("Unintended Usage"). Unintended Usage include atomic energy control instruments, airplane or spaceship instruments, transportation instruments, traffic signal instruments, combustion control instruments, medical instruments, all types of safety devices, etc.. Unintended Usage of Semiconductor products listed in this document shall be made at the customer's own risk.